

MATH 999, Summer 2026
Note 3: Galois Correspondence
Trevor Hyde

“Would you tell me, please, which way I ought to go from here?”
“That depends a good deal on where you want to get to,” said the Cat.
“I don’t much care where...”
“Then it doesn’t matter which way you go.”

From *Alice in Wonderland* by Lewis Carroll.

In the previous note we defined Galois extensions and Galois groups. Our next goal is to study intermediate fields. That is, given a field extension L/K , what are all the fields F which satisfy $K \subseteq F \subseteq L$? We first study this question for Galois extensions, which you should think of as the *very nicest* of all finite extensions.

Problem 3.1. Suppose that L/K is a Galois extension. If $K \subseteq F \subseteq L$ is an intermediate field, prove that L/F is also a Galois extension and that $\text{Gal}(L/F) \subseteq \text{Gal}(L/K)$ is a subgroup.

Let $\text{Int}(L/K) := \{F : K \subseteq F \subseteq L\}$ be the set of all intermediate fields and let $\text{Sub}(L/K) = \{H : H \subseteq \text{Gal}(L/K)\}$ denote the set of all subgroups of $\text{Gal}(L/K)$. Then Problem 3.1 shows that for a Galois extension L/K we have a function $\Psi : \text{Int}(L/K) \rightarrow \text{Sub}(L/K)$ defined by $\Psi(F) := \text{Gal}(L/F)$. The Galois Correspondence, which we are working towards, asserts that Ψ is a bijection for all Galois extensions L/K . That is, the intermediate fields of L/K correspond precisely to subgroups of $\text{Gal}(L/K)$.

The best way to prove that a function is a bijection is to find its inverse. So that means we should start with a subgroup $H \subseteq \text{Gal}(L/K)$ and we want to use this to produce an intermediate field of L/K . We do this by constructing the *fixed field* of H .

Problem 3.2. Let L/K be a Galois extension and let $H \subseteq \text{Gal}(L/K)$ be a subgroup. Let

$$L^H := \{\alpha \in L : \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}.$$

Prove that L^H is a field and that $K \subseteq L^H \subseteq L$. Hence we have a function $\Phi : \text{Sub}(L/K) \rightarrow \text{Int}(L/K)$ defined by $\Phi(H) := L^H$. If $H_1 \subseteq H_2 \subseteq \text{Gal}(L/K)$ are subgroups, prove that $L^{H_2} \subseteq L^{H_1}$.

Now we have a function Φ mapping in the opposite direction as Ψ . We want to show that Φ and Ψ are inverses. To show this we need to prove that $\Psi\Phi(H) = H$ and $\Phi\Psi(F) = F$ for all $H \in \text{Sub}(L/K)$ and all $F \in \text{Int}(L/K)$. Let’s unpack this first equality. By definition we have $\Psi\Phi(H) = \text{Gal}(L/L^H)$. Then our goal is to show that $\text{Gal}(L/L^H) = H$. One direction is straightforward.

Problem 3.3. Prove that for all $H \in \text{Sub}(L/K)$ we have $H \subseteq \text{Gal}(L/L^H)$.

The other direction is more challenging. We need to show that the *only* automorphisms of L which fix L^H are the elements of H .

Theorem 3.1 (Artin). For all $H \in \text{Sub}(L/K)$ we have $[L : L^H] \leq |H|$.

Proof. Let $H = \{\sigma_1, \dots, \sigma_n\}$, so that $n = |H|$. Suppose for sake of contradiction that $[L : L^H] > n$. Then we can find at least $n + 1$ elements $\alpha_0, \dots, \alpha_n \in L$ which are linearly independent over L^H . Let $x_0, \dots, x_n$ be variables and consider the following system of n linear equations:

$$\sigma_i(\alpha_0)x_0 + \sigma_i(\alpha_1)x_1 + \dots + \sigma_i(\alpha_n)x_n = 0,$$

for $1 \leq i \leq n$. Since this system has more variables than equations, it must have a nonzero solution. Suppose that $(\beta_0, \dots, \beta_n)$ is a solution with the fewest number of nonzero β_j . Without loss of generality, we may relabel in order to assume that $\beta_0 \neq 0$. Dividing everything by β_0 we may also assume without loss of generality that $\beta_0 = 1$.

Suppose that σ_1 is the identity. If all the β_i were elements of L^H , then

$$\alpha_0 + \alpha_1\beta_1 + \dots + \alpha_n\beta_n = 0.$$

However, we assumed that the α_i are linearly independent over L^H . Thus there must be some $\beta_j \notin L^H$. Hence there is some element $\sigma_k \in H$ which does not fix β_j , so $\sigma_k(\beta_j) \neq \beta_j$. We now apply σ_k to our entire system of equations with the β solution plugged in to get equations

$$\sigma_k\sigma_i(\alpha_0) + \sigma_k\sigma_i(\alpha_1)\sigma_k(\beta_1) + \dots + \sigma_k\sigma_i(\alpha_n)\sigma_k(\beta_n) = \sigma_k(0) = 0.$$

Since H is a subgroup and $\sigma_k \in H$, it follows that $\sigma_k\sigma_i \in H$ for each i . Hence we can write $\sigma_{i'} := \sigma_k\sigma_i$. [Prove that the map \$\sigma_i \mapsto \sigma_{i'}\$ is a bijection of \$H\$ with itself.](#) Thus we have

$$\sigma_i(\alpha_0) + \sigma_i(\alpha_1)\sigma_k(\beta_1) + \dots + \sigma_i(\alpha_n)\sigma_k(\beta_n) = 0$$

for each $1 \leq i \leq n$. Taking the difference between these equations and our original equations we get

$$\sigma_i(\alpha_1)(\sigma_k(\beta_1) - \beta_1) + \dots + \sigma_i(\alpha_n)(\sigma_k(\beta_n) - \beta_n) = 0.$$

Notice that the α_0 terms cancel in the difference. Thus $(x_0, \dots, x_n) = (0, \sigma_k(\beta_1) - \beta_1, \dots, \sigma_k(\beta_n) - \beta_n)$ is a solution of our original system of equations. It must be a nonzero solution since $\sigma_k(\beta_j) - \beta_j \neq 0$ by construction. Furthermore, this solution has strictly fewer nonzero terms than $(x_0, \dots, x_n) = (\beta_0, \dots, \beta_n)$. But this contradicts our assumption that the β solution had the fewest nonzero coordinates. Therefore, we must have $[L : L^H] \leq n$ after all. $\square$

Combining Artin's theorem with a result from the previous note we have

$$|H| \leq |\text{Gal}(L/L^H)| = [L : L^H] \leq |H|.$$

Therefore $|\text{Gal}(L/L^H)| = |H|$ and $H \subseteq \text{Gal}(L/L^H)$, which shows that $\text{Gal}(L/L^H) = H$. We have finished proving the hardest part of the Galois correspondence!

Next we turn to proving that $\Phi\Psi(F) = F$. Unpacking definitions, this is equivalent to showing that $L^{\text{Gal}(L/F)} = F$.

Problem 3.4. [Prove that \$F \subseteq L^{\text{Gal}\(L/F\)}\$ for all \$F \in \text{Int}\(L/K\)\$.](#) To finish the proof, we just need to combine pieces we already have. [Prove that \$\[L : L^{\text{Gal}\(L/F\)}\] = \[L : F\]\$ and use this to conclude that \$F = L^{\text{Gal}\(L/F\)}\$.](#)

We have now proved the following fundamental result of Galois theory.

Theorem 3.2 (Galois Correspondence, Part I). Let L/K be a Galois extension. Then there is a bijection between $\text{Sub}(L/K)$, the subgroups of $\text{Gal}(L/K)$, and $\text{Int}(L/K)$ the intermediate fields $K \subseteq F \subseteq L$. The bijection is given by $F \mapsto \text{Gal}(L/F)$ with inverse $H \mapsto L^H$.

So what does the Galois correspondence tell us? It essentially tells us that the structure of a Galois extension L/K is entirely determined by the structure of its Galois group $\text{Gal}(L/K)$. For one, the intermediate fields $K \subseteq F \subseteq L$ exactly correspond to the subgroups of $\text{Gal}(L/K)$. Thus if we know the group $\text{Gal}(L/K)$, we can use group theory to understand all its subgroups, and then the Galois correspondence translates this into a complete picture of all the intermediate fields.

To start to get a feel for how this works, we analyze an example.

Problem 3.5. Consider the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$. First we need to show that this extension is Galois.

1. Prove that $x^2 - 3$ does not have any roots in $\mathbb{Q}(\sqrt{2})$ and use this to prove that

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

2. Prove that $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is the splitting extension of the separable polynomial

$$f(x) = (x^2 - 2)(x^2 - 3).$$

Hence $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is a Galois extension. Note that this involves two steps: you need to show that $f(x)$ splits completely in this extension, and then you need to argue that it is the smallest extension over which our polynomial splits.

We now know that $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is a degree 4 Galois extension. Let $G := \text{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$. Then we know that $|G| = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$. Up to isomorphism, there are exactly two groups of order 4, namely the cyclic group C_4 and the product $C_2 \times C_2$ of two cyclic groups of order 2. So which one is G isomorphic to?

3. Let $\sigma_2 \in \text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ be the automorphism defined by $\sigma_2(\sqrt{2}) = -\sqrt{2}$. Use Lemma 2.3 to prove that σ_2 has a unique extension to an element $\sigma_2 \in G$ such that $\sigma_2(\sqrt{3}) = \sqrt{3}$. Make a similar argument to show there is an element $\sigma_3 \in G$ such that $\sigma_3(\sqrt{3}) = -\sqrt{3}$ and $\sigma_3(\sqrt{2}) = \sqrt{2}$.
4. Prove that $G \cong C_2 \times C_2$.
5. The group G has a total of five subgroups. Find all of the subgroups (give generators for each one) and determine their order.
6. For each subgroup $H \subseteq G$, find generators for the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})^H$. Most of these are straight forward, but one may be tricky! The Galois correspondence tells us these are all of the intermediate fields of the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$.
7. Which of the five possible intermediate fields you found in the previous problem is $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ equal to? **Hint:** Use the Galois group!